

# Network design practice 5-1

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```
Scanning dependencies of target scratch_first
[ 0%] Building CXX object scratch/CMakeFiles/scratch_first.dir/first.cc.o
[ 0%] Linking CXX executable ../../build/scratch/ns3.42-first-de
fault
At time +2s client sent 1024 bytes to 10.1.1.2 port 91963
At time +2.00369s server received 1024 bytes from 10.1.1.1 port 4
9153
At time +2.00369s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 9
```

Figure 1: successful build of the code

Successfully run first.cc

```
Throughput: 13.2703 Mbit/s
ns3ubuntu:~/ns-0-RAN-FluxrC/nmwave-LENA-ora$
ns3ubuntu:~/ns-0-RAN-FluxrC/nmwave-LENA-ora$
ns3ubuntu:~/ns-0-RAN-FluxrC/nmwave-LENA-ora$ ./ns3 run scratch/connet-5-1-example.cc -- -nMpdus=4 --payloadSize=600
Throughput: 33.252 Mbit/s
ns3ubuntu:~/ns-0-RAN-FluxrC/nmwave-LENA-ora$ ./ns3 run scratch/connet-5-1-example.cc -- -nMpdus=2 --payloadSize=600
Throughput: 23.2728 Mbit/s
```

Figure 2: Variation on payloadSize, nMpdus, and enableRts

Run code with payloadsize 100, nMpdus 4, and the throughput is 13.27Mbit/s. With increased payload size 600, throughput is 33.25Mbit/s. Portion of data is increased when using a larger payload size. With nMpdus set to 2, throughput is decreased to 23.27Mbit/s. This is because large portion of data is transferred when using larger mpdus size.

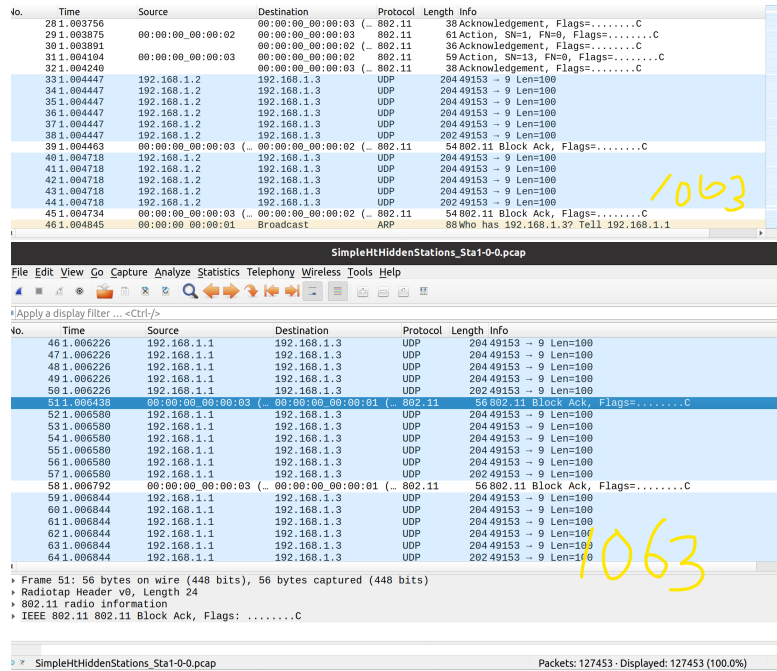


Figure 3: Wireshark at AP(top) and STA(bottom) with no CTS enabled

We observe only block ACK is controlling the whole traffic, and no Clear-To-Send signal available in the first three practices.

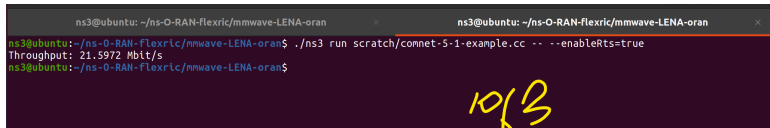


Figure 4: throughput with CTS enabled

Throughput with CTS enabled 22Mbit/s is observed. Since RTS/CTS signal consumes a portion of the available bandwidth, the total throughput payload capacity.

|            |                      |                      |        |                                          |
|------------|----------------------|----------------------|--------|------------------------------------------|
| 521.005063 | 00:00:00_00:00:02 (- | 00:00:00_00:00:03 (- | 802.11 | 47 Request-to-send, Flags=.....C         |
| 531.005079 |                      | 00:00:00_00:00:02 (- | 802.11 | 36 Clear-to-send, Flags=.....C           |
| 541.005367 | 192.168.1.2          | 192.168.1.3          | UDP    | 1565 49153 -> 9 Len=1472                 |
| 551.005383 |                      | 00:00:00_00:00:02 (- | 802.11 | 36 Acknowledgement, Flags=.....C         |
| 561.005554 | 00:00:00_00:00:02 (- | 00:00:00_00:00:03 (- | 802.11 | 47 Request-to-send, Flags=.....C         |
| 571.005570 |                      | 00:00:00_00:00:02 (- | 802.11 | 36 Clear-to-send, Flags=.....C           |
| 581.005558 | 192.168.1.2          | 192.168.1.3          | UDP    | 1565 49153 -> 9 Len=1472                 |
| 591.005874 |                      | 00:00:00_00:00:02 (- | 802.11 | 36 Acknowledgement, Flags=.....C         |
| 601.005981 | 00:00:00_00:00:01    | Broadcast            | ARP    | 88 who has 192.168.1.3? Tell 192.168.1.1 |
| 611.006217 | 00:00:00_00:00:03 (- | 00:00:00_00:00:01 (- | 802.11 | 45 Request-to-send, Flags=.....C         |
| 621.006341 |                      | 00:00:00_00:00:03 (- | 802.11 | 38 Clear-to-send, Flags=.....C           |

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| No.            | Time | Source               | Destination          | Protocol | Length                           | Info |
|----------------|------|----------------------|----------------------|----------|----------------------------------|------|
| 43757.6.825656 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Clear-to-send, Flags=.....C   |      |
| 43758.6.825344 |      | 192.168.1.2          | 192.168.1.3          | UDP      | 1565 49153 -> 9 Len=1472         |      |
| 43759.6.825360 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Acknowledgement, Flags=.....C |      |
| 43760.6.825495 |      | 00:00:00_00:00:02 (- | 00:00:00_00:00:03 (- | 802.11   | 47 Request-to-send, Flags=.....C |      |
| 43761.6.825511 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Clear-to-send, Flags=.....C   |      |
| 43762.6.825799 |      | 192.168.1.2          | 192.168.1.3          | UDP      | 1565 49153 -> 9 Len=1472         |      |
| 43763.6.825815 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Acknowledgement, Flags=.....C |      |
| 43764.6.826013 |      | 00:00:00_00:00:02 (- | 00:00:00_00:00:03 (- | 802.11   | 47 Request-to-send, Flags=.....C |      |
| 43765.6.826029 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Clear-to-send, Flags=.....C   |      |
| 43766.6.826318 |      | 192.168.1.2          | 192.168.1.3          | UDP      | 1565 49153 -> 9 Len=1472         |      |
| 43767.6.826334 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Acknowledgement, Flags=.....C |      |
| 43768.6.826595 |      | 00:00:00_00:00:02 (- | 00:00:00_00:00:03 (- | 802.11   | 47 Request-to-send, Flags=.....C |      |
| 43769.6.826611 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Clear-to-send, Flags=.....C   |      |
| 43770.6.826899 |      | 192.168.1.2          | 192.168.1.2          | UDP      | 1565 49153 -> 9 Len=1472         |      |
| 43771.6.826915 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Acknowledgement, Flags=.....C |      |
| 43772.6.827167 |      | 00:00:00_00:00:02 (- | 00:00:00_00:00:03 (- | 802.11   | 47 Request-to-send, Flags=.....C |      |
| 43773.6.827183 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Clear-to-send, Flags=.....C   |      |
| 43774.6.827471 |      | 192.168.1.2          | 192.168.1.3          | UDP      | 1565 49153 -> 9 Len=1472         |      |
| 43775.6.827487 |      |                      | 00:00:00_00:00:02 (- | 802.11   | 36 Acknowledgement, Flags=.....C |      |

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Figure 5: Wireshark at AP(top) and STA(bottom) with CTS enabled

Wireshark at AP and STA after CTS signal is enabled. Notice compared to the previous wireshark report CTS signal is visible. Stations are allowed to send only after ACK-RTS-CTS process is done.